

Roll No.....

End Semester Examination

Name of the Program: B.Tech(ECE)
Name of the Course: Signals & Systems

Semester: III
Course Code: TEC 304

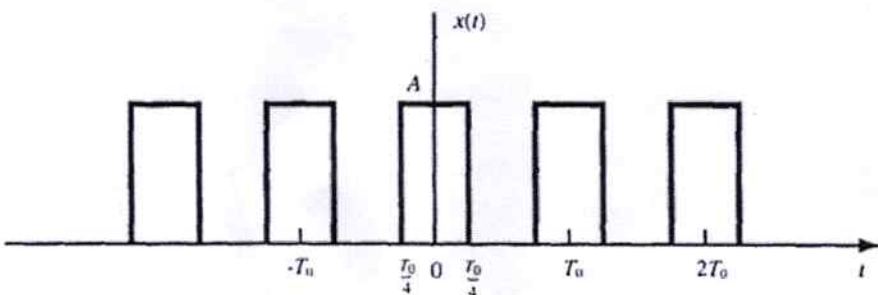
Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Determine the power and the rms value of the given signals (i) $x(t) = 5 \cos\left(50t + \frac{2\pi}{3}\right)$ (ii) $x(t) = 10 \sin\left(50t + \frac{\pi}{4}\right) + 16 \cos\left(50t + \frac{\pi}{3}\right)$ (iii) $x[n] = 2e^{j3n}$.	CO1
(b)	Show that the following properties hold for the derivative of $\delta(t)$: (a) $\int_{-\infty}^{\infty} \phi(t) \delta'(t) dt = -\phi'(0)$ where $\phi'(0) = \left. \frac{d\phi(t)}{dt} \right _{t=0}$ (b) $t \delta'(t) = -\delta(t)$	
(c)	Determine the even and odd components of the following signals: (i) $x(t) = u(t)$ (ii) $x[n] = u[n]$.	
Q2	(20 marks)	
(a)	The input $x(t)$ and the impulse response $h(t)$ of a continuous time LTI system are given by $x(t) = u(t)$ and $h(t) = e^{-\alpha t} u(t)$, $\alpha > 0$. Compute the output $y(t)$.	CO 2
(b)	Compute $y[n] = x[n] * h[n]$, where $x[n] = \alpha^n u[n]$, $h[n] = \beta^n u[n]$.	
(c)	Verify the associative property, that is, $\{x[n] * h_1[n]\} * h_2[n] = x[n] * \{h_1[n] * h_2[n]\}$	
Q3	(20 marks)	
(a)	Discuss the complex conjugate property of the Fourier series.	CO3
(b)	Determine the complex exponential Fourier series representation for each of the following signals: (a) $x(t) = \cos(\omega_0 t)$ (b) $x(t) = \cos\left(2t + \frac{\pi}{4}\right)$	
(c)	Consider the periodic square wave $x(t)$ shown in Fig. (a) Determine the complex exponential Fourier series of $x(t)$. (b) Determine the trigonometric Fourier series of $x(t)$.	

		
Q4	(20 marks)	
(a)	Find the Fourier transform of the following signals (i) $x(t) = \delta(t - t_0)$ (ii) $x(t) = u(t)$ (iii) $x(t) = \tau \sin c\left(\frac{t\tau}{2}\right)$	CO4
(b)	Verify Parseval's theorem for the Fourier Transform.	
(c)	Verify the integration property, that is, $\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{F.T} \pi X(0) \delta(\omega) + \frac{1}{j\omega} X(\omega)$	
Q5	(20 marks)	
(a)	Find Laplace transform of the following signals (i) $x(t) = e^{-(t+1)} u(t+1)$ (ii) $x(t) = -t e^{-2t} u(t)$	CO5 & CO6
(b)	Verify the time-scaling property $x(at) \leftrightarrow \frac{1}{ a } X\left(\frac{s}{a}\right) \quad R' = aR$	
(c)	Derive the relationship between the Fourier Transform and the Laplace Transform and discuss their regions of convergence.	